

KIHYUN YU

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<https://kihyun-yu.github.io>

RESEARCH INTERESTS

My research focuses on **safe reinforcement learning (RL) theory**. In particular, I design algorithms for **constrained MDPs** with provable guarantees under various settings; adversarial, non-stationary, function approximation, and infinite-horizon formulations. I am also broadly interested in **sequential decision-making**, including **bandit algorithms**, **online convex optimization**, and their real-world applications.

EDUCATION

- KOREA ADVANCED INSTITUTE OF SCIENCE AND TECHNOLOGY (KAIST) South Korea
 - Integrated M.S.–Ph.D. Program in Industrial and Systems Engineering 03/2024 – present
 - Advisor: [Prof. Dabeen Lee](#)
 - GPA: 4.2 / 4.3
- GWANGJU INSTITUTE OF SCIENCE AND TECHNOLOGY (GIST) South Korea
 - B.S. in Electrical Engineering and Computer Science 03/2018 – 02/2024
 - Minor in Mathematics
 - GPA: 4.08 / 4.5 (*cum laude*)
 - Two-year leave for mandatory military service (2020 – 2021)
- SEJONG SCIENCE HIGH SCHOOL South Korea
 - 03/2015 – 02/2018

PUBLICATIONS

PREPRINTS

- [1] **Kihyun Yu**, Junehee Lee, Dabeen Lee. Constrained Online Convex Optimization without Slater’s Condition. Available at: <https://arxiv.org/abs/2606.31480>
- [2] **Kihyun Yu**, Beomhan Baek, Dabeen Lee. Learning Weakly Communicating Average-Reward CMDPs: Strong Duality and Improved Regret. Available at: <https://arxiv.org/abs/2605.11586>

PUBLICATIONS

- [3] **Kihyun Yu**, Seoungbin Bae, Dabeen Lee. Near-Optimal Primal-Dual Algorithm for Learning Linear Mixture CMDPs with Adversarial Rewards. *IEEE Conference on Decision and Control (CDC)*, 2026. Available at: <https://arxiv.org/abs/2603.27884>
- [4] **Kihyun Yu**, Seoungbin Bae, Dabeen Lee. Primal-Dual Policy Optimization for Linear CMDPs with Adversarial Losses. *International Conference on Learning Representations (ICLR)*, 2026. Available at: <https://arxiv.org/abs/2605.11535>
- [5] Jiahui Zhu, **Kihyun Yu**, Dabeen Lee, Xin Liu, Honghao Wei. An Optimistic Algorithm for online CMDPs with Anytime Adversarial Constraints. *International Conference on Machine Learning (ICML)*, 2025. Available at: <https://arxiv.org/abs/2505.21841>
- [6] **Kihyun Yu**, Duksang Lee, William Overman, and Dabeen Lee. Improved Regret Bound for Safe Reinforcement Learning via Tighter Cost Pessimism and Reward Optimism. *Reinforcement Learning Conference (RLC)*, 2025. Available at: <https://rlj.cs.umass.edu/2025/papers/Paper41.html>

ACADEMIC SERVICES

- Conference Reviewer: ICML 2026 ([silver reviewer](#)), NeurIPS 2026, IEEE CDC 2026, AAAI 2027
- Journal Reviewer: TMLR

TEACHING EXPERIENCE

- Teaching Assistant, Operations Research: Optimization (KAIST) 03/2024 – 06/2024
- Teaching Assistant, Differential Equations (GIST) 09/2023 – 12/2023
- Teaching Assistant, Single Variable Calculus (GIST) 03/2023 – 06/2023

HONORS

- Scholarship for Academic Excellence
 - Awarded for 6 semesters (Spring 2018, Fall 2018, Spring 2019, Fall 2019, Spring 2022, Spring 2023)

EXPERIENCE

- Information Processing, Controlling, and Network Lab (GIST) South Korea
 - Undergraduate Research Intern (Advisor: Prof. Heung-No Lee) 01/2023 – 12/2023
 - Conducted research on lattice-based cryptography
- Artificial Intelligence Lab (GIST) South Korea
 - Undergraduate Research Intern (Advisor: Prof. Kyoobin Lee) 07/2022 – 12/2022
 - Conducted research on robust image classification under noisy labels
- University of Cambridge United Kingdom
 - Summer Session Programme (Machine Learning) 07/2019 – 08/2019

LANGUAGES AND TECHNICAL SKILLS

- Korean (native), English (conversational)
- C/C++, Python, PyTorch, $\LaTeX$

OTHER ACTIVITIES

- Volunteer Math Mentor 03/2018 – 12/2018
 - Provided small-group math mentoring for middle and high school students